UIDAI

DATA HACKATHON

2026

TEAM MEMBERS DETAIL

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Introduction to Raw Data

ENROLLMENT DATASET

The Enrollment dataset captures the entry point of a citizen into the Aadhaar system. It records the creation of new Aadhaar numbers across different age groups.

Columns = [“date”, ”state”, “district”, “pincode”, “age_0_5”, “age_5_17",”age_18_greater”]

Number Of CSV File Given : 3

Original Number of Rows : 1006029

DEMOGRAPHIC DATASET

The Demographic dataset records changes to non-biometric information. In a mobile and evolving population, this is the most frequently used service.

Columns = [“date”, ”state”, “district”, “pincode”, “bio_age_5_17, “bio_age_17_"]

Number of CSV File Given : 5

Original Number of Rows : 2071700

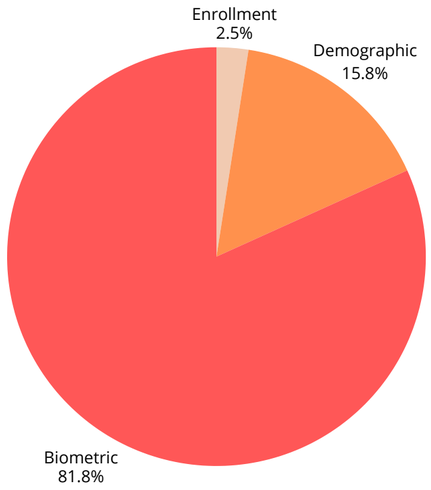
BIOMETRIC UPDATE DATASET

Biometric data is the "security layer" of Aadhaar. This dataset tracks when citizens update their fingerprints, iris scans, or facial photographs.

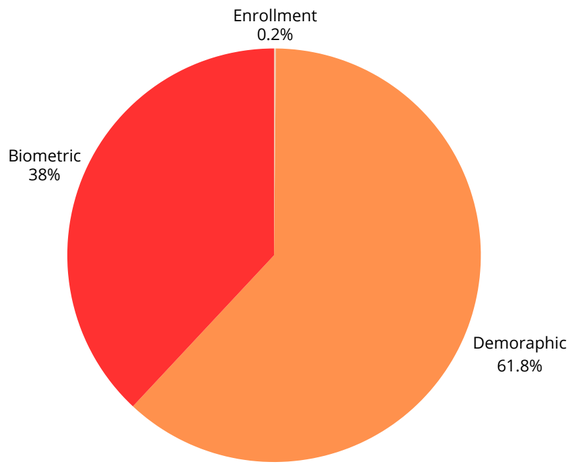
Columns = [“date”, ”state”, “district”, “pincode”, “demo_age_5_17”, “demo_age_17_"]

Number of CSV File Given : 4

Original Number of Rows : 1861108



5-17 Age Group



Adult(17+) Age Group

Resolving Ambiguities & Anomalies

Bridging the Data Gap

1. Administrative Alignment (The UT Merger)
Ambiguity: Fragmented entries for Daman, Diu, and Dadra & Nagar Haveli.
Action: Unified 4 naming variants into one official administrative unit.
Result: Accurate, consolidated reporting for the merged Union Territory.

2. Schema Harmonization (Age Buckets)
Ambiguity: Inconsistent "Adult" definitions (17+ vs. 18+) across files.
Action: Merged into a unified "Adult Transaction" category.
Result: Enabled 100% comparable metrics for Enrolment vs. Lifecycle Updates.

3. Regional Standardization (State/District)
Ambiguity: Over 15 spelling variations per state (e.g., "West Bangal", "Orissa").
Action: Automated mapping to official Indian Census nomenclature.
Result: Eliminated statistical dilution; ensured data totals are 100% accurate.

5. Redundancy Removal (Duplicate Rows)
Anomaly: Found 94k+ duplicate rows in Biometric Dataset , 22k+ duplicate records in Enrollment Dataset and 473k duplicate records in Demographic Dataset
Action: Implemented strict multi-column deduplication during the ingestion phase.
Result: Prevented artificial inflation of activity; ensured "True Transaction" counts.

4. Technical Hygiene
Ambiguity: Mismatched date formats and Pincodes stored as decimals.
Action: Forced global datetime standard and Integer type-casting.
Result: Enabled high-performance "Master Merge" across millions of rows.

Resolving Ambiguities & Anomalies

From Raw Data to Master Dataset

Raw Data Collection
Gathering raw data from various sources

Identify Data Issues
Recognizing issues like duplicate rows and inconsistent formats

State Name Normalization
Standardizing state names using mapping and replacement

Duplicate Row Removal
Eliminating duplicate rows using the drop_duplicates() function

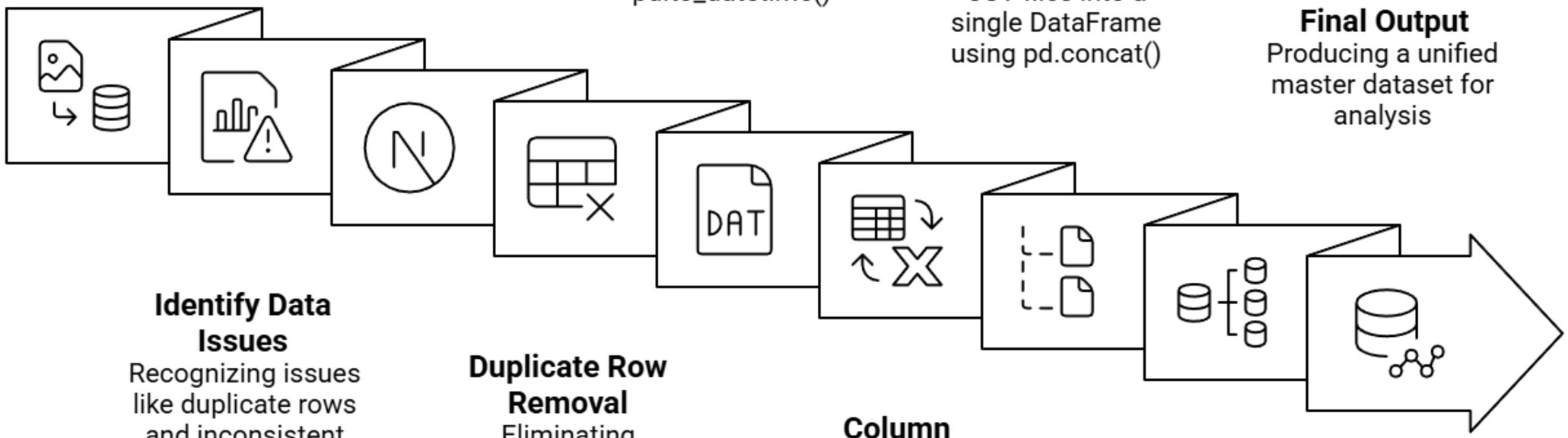
Date Parsing
Converting date strings to datetime objects using pd.to_datetime()

Column Renaming
Renaming columns to a standardized format using rename()

File Consolidation
Merging multiple CSV files into a single DataFrame using pd.concat()

Dataset Join
Combining datasets using an outer join to create a master dataset

Final Output
Producing a unified master dataset for analysis



Problem Statement

The Challenge: Navigating the "Maintenance Era" of Digital Identity

- **Infrastructure Blind Spots:** High Demographic activity paired with Zero Biometric updates reveals localized hardware failures and "Biometric Blackholes".
- **The Compliance Gap:** While infant enrollment (0-5) is successful, mandatory biometric updates for children (5-17) are lagging, creating future authentication risks.
- **Service Pressure Imbalance:** Operations are over-stressed in migration hubs (e.g., Bengaluru) while rural stations remain "underperforming" or "dead" due to market saturation.

Systematic Solution

The "Adaptive Resource" & SPI-Driven Model

SPI-Driven Optimization: Transition from static staffing to Service Pressure Index (SPI) planning. Deploy resources where workload-to-capacity ratios are highest.

Human Resource Redistribution: Instead of closing "Saturated" centers, reduce their permanent staff to a skeleton crew and pivot them to Self-Service Support.

Dynamic Response Teams: Reassign surplus staff from low-SPI rural zones to "High-Pressure" urban migration hubs (e.g., Bengaluru, Thane) to handle surges.

Targeted Hardware Audits: Use the Anomaly Score to identify and fix "Biometric Blackholes" where centers are active but scanners are non-functional.

Why It Works

Operational Flexibility & Efficiency

Eliminates "Dead Cost": Reduces the high cost of maintaining fully staffed centers in 100% saturated markets.

Reduces Citizen Wait-Time: Shifting HR to high-pressure districts (like Pune or Surat) directly improves service delivery speed.

Maintains Local Presence: Ensures rural citizens still have access points, preventing "Identity Exclusion" in remote areas.

Success Metrics: Measuring Operational Excellence

SPI Equilibrium (Load Balance)

- **Target:** Maintain a Service Pressure Index (SPI) between 0.8 and 1.2 across all centers.
- **Success Indicator:** Elimination of "Over-stressed" urban hubs and "Idle" saturated rural stations through the dynamic redistribution of staff.

Hardware Uptime & "Anomaly Score" Recovery

- **Target:** Reduce the "Biometric Blackhole" count by 90% within the first 90 days.
- **Success Indicator:** A verified drop in districts showing a high Demographic-to-Biometric Ratio on the Geospatial Audit map.

Mandatory Compliance Rate (MBU)

- **Target:** Achieve a 25% increase in mandatory biometric updates for the 5–17 age group.
- **Success Indicator:** Narrowing the gap between infant enrollment (0-5) and subsequent compliancedriven visits.

HR Efficiency & Wait-Time Reduction

- **Target:** A 30% reduction in average citizen wait-times in high-migration districts.
- **Success Indicator:** Increased "Transactions Per Staff Member" following the strategic move of personnel from saturated markets to urban magnets.

Aadhaar Enrolment Analysis

Executive Summary

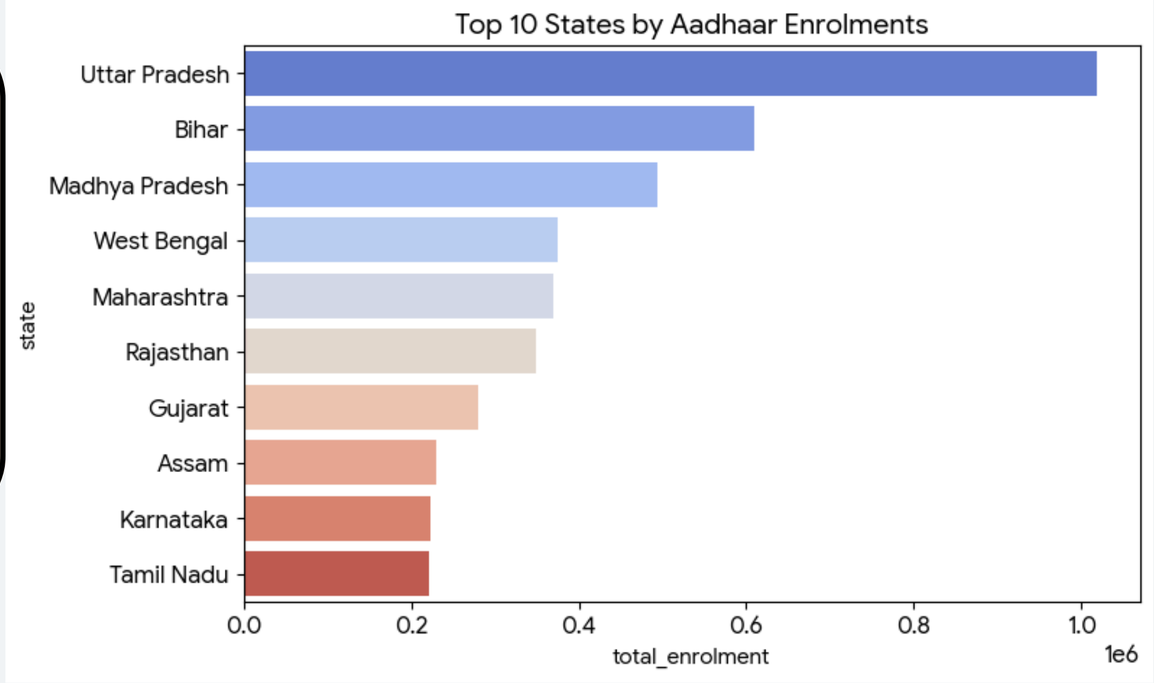
- Total Enrolments: **5.33M (approximately)**
- New Born/Infant Enrolments (0-5): **3.47M (approximately)**
- Leading State by Total Enrollments : **Uttar Pradesh**
- Leading District by Total Enrollments : **Thane**
- Data Scope: National Enrolment Centers **(Mar-Dec 2025)**

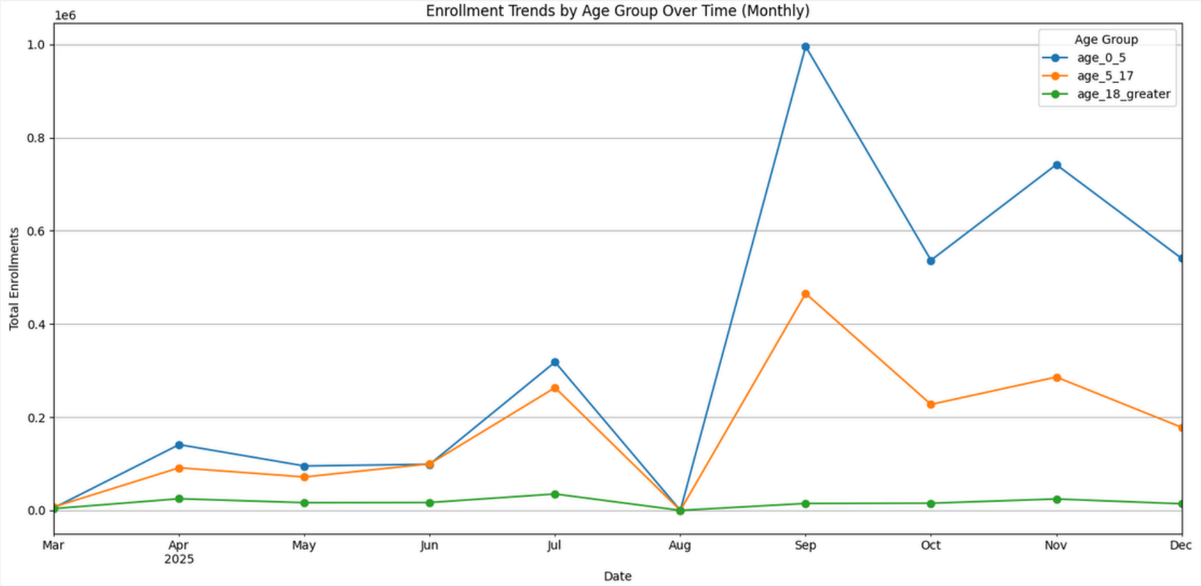
Enrollment Analysis

Tracking the Digital Onboarding

Regional Growth Leaders

- **High-Volume States:** Identify the top 3 states (e.g., Uttar Pradesh, Bihar, West Bengal) where absolute enrollment remains high due to population density.
- **The "Catch-up" Zones:** High adult enrollment in specific rural districts suggests these were previously underserved areas where digital identity is now successfully reaching the "Last Mile."

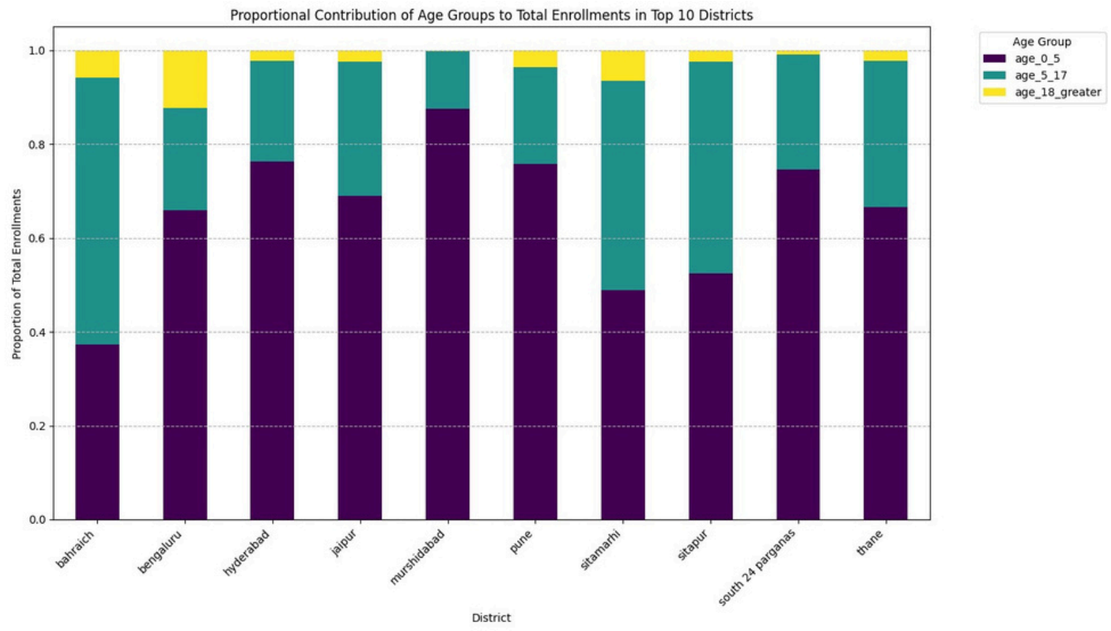




Impact of Outreach Camps

Pattern: Visible surges in 5-17 enrollment during specific months (correlating with the start of the academic year).

Solution: Proactive deployment of mobile enrollment vans to schools during June-July based on historical trend lines.



- **Baal Aadhaar (0-5):** Primary growth driver; successfully integrated with birth registrations.
- **School-Age (5-17):** Steady intake; fueled by school admissions and scholarship linkages.
- **Adult Enrollment (18+):** Declining trend; confirms near 100% saturation in urban areas.

The Data: While rural districts show negligible 18+ enrollment, Bengaluru shows a significant volume (over 7,000+ in our sample), which is higher than most other districts.

The Reason:

- **National Trend:** Adult enrollment (18+) is hitting a "Saturation Floor" (near 0 in rural districts).
- **The Outlier:** Bengaluru shows 3 times higher adult enrollment than rural benchmarks.
- **The Driver**: High Economic Migration; transient workforce enrolling as adults for city-specific benefits and corporate documentation.
- **Strategic Takeaway:** Maintain high-capacity adult enrollment hubs in Urban Migration Magnets, even while phasing them out in saturated rural zones.

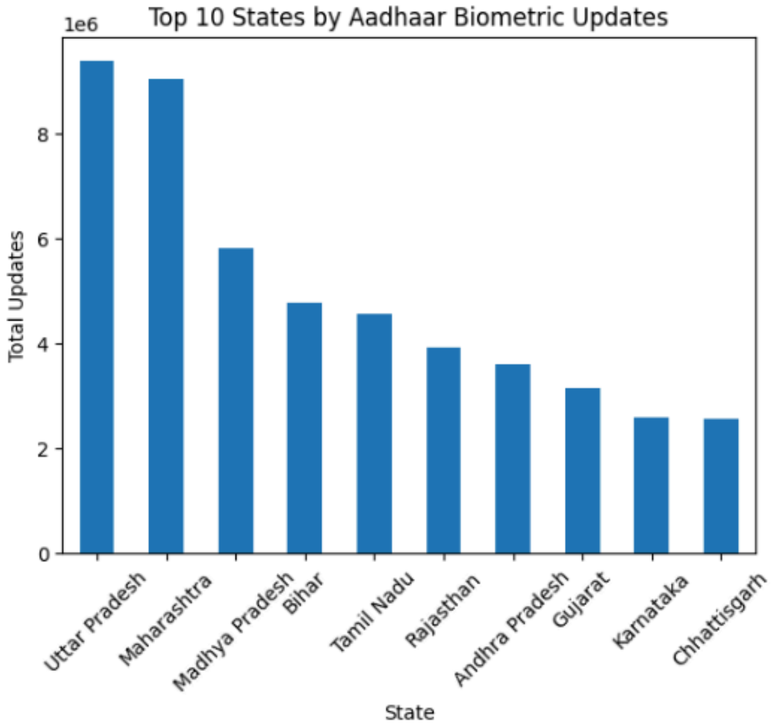
Aadhaar Biometric Analysis

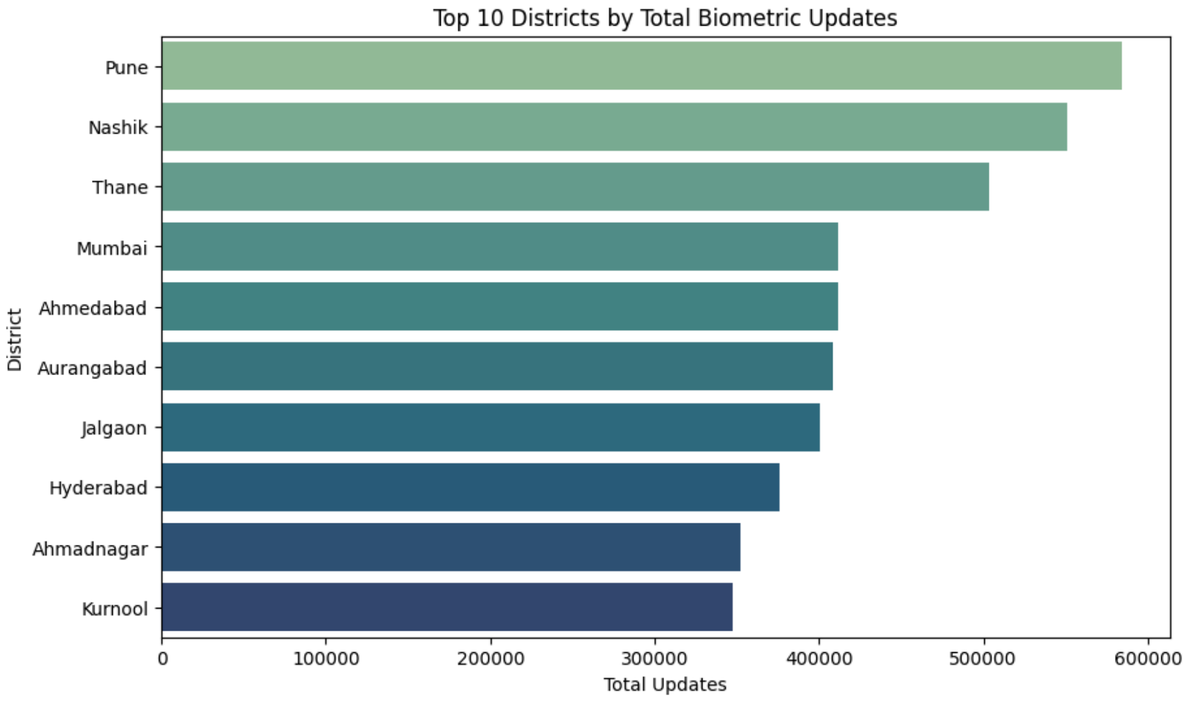
Executive Summary

- Total Updates: **68.21M (approximately)**
- Primary Contributors: **Adults (18+)** with significant child updates (5–17)
- Leading State: **Uttar Pradesh**
- Leading District: **Sitapur**
- Data Scope: National Enrolment Centers **(Mar-Dec 2025)**

Biometric Analysis

Regional Load & Growth Leaders





- Migration hubs and service economies exhibit persistent update pressure
- Rural districts show low but non-zero demand, indicating compliance-driven visits

- High biometric update volumes are concentrated in urban and semi-urban districts

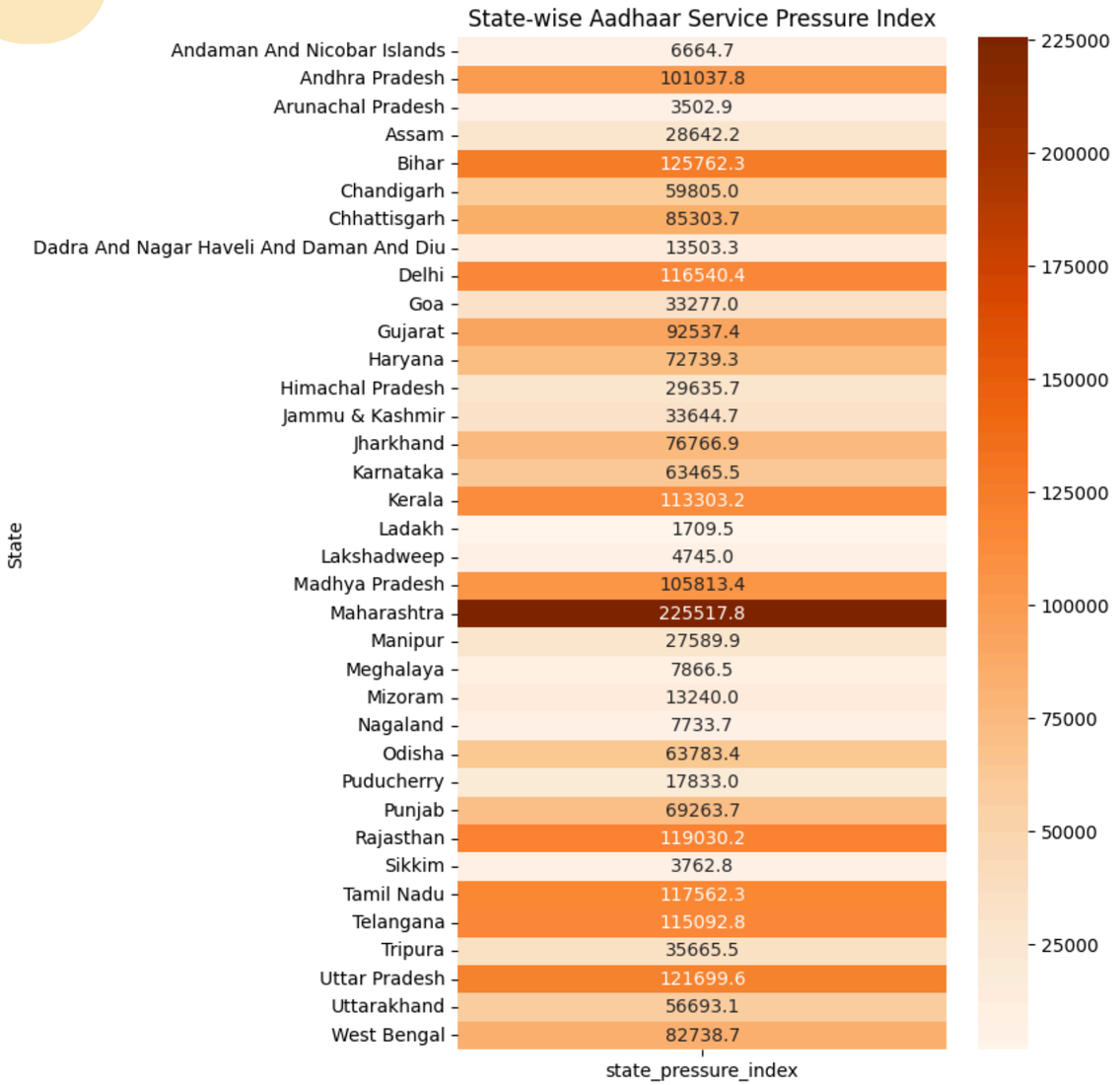
Biometric Analysis

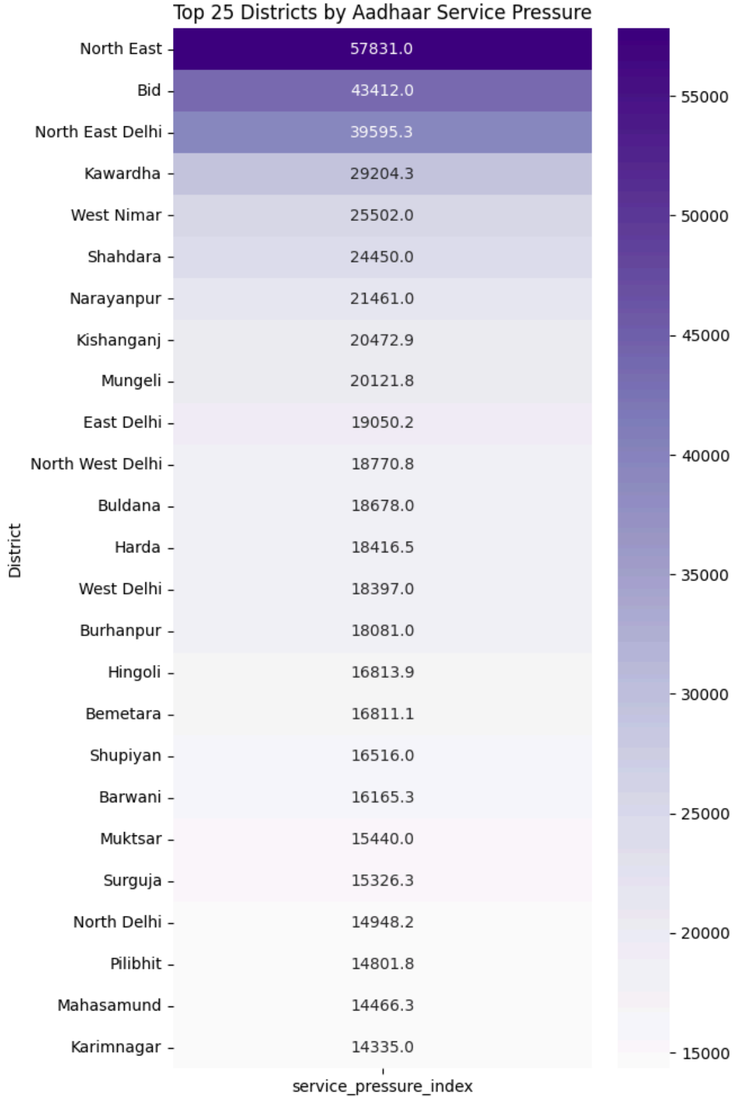
Service Pressure & Load Concentration

- A small number of districts handle a disproportionate share of biometric traffic.
- Indicates operational stress rather than population effect.
- Validates need for SPI (Service Pressure Index)–based planning.

Biometric Analysis

Service Pressure & Load Concentration





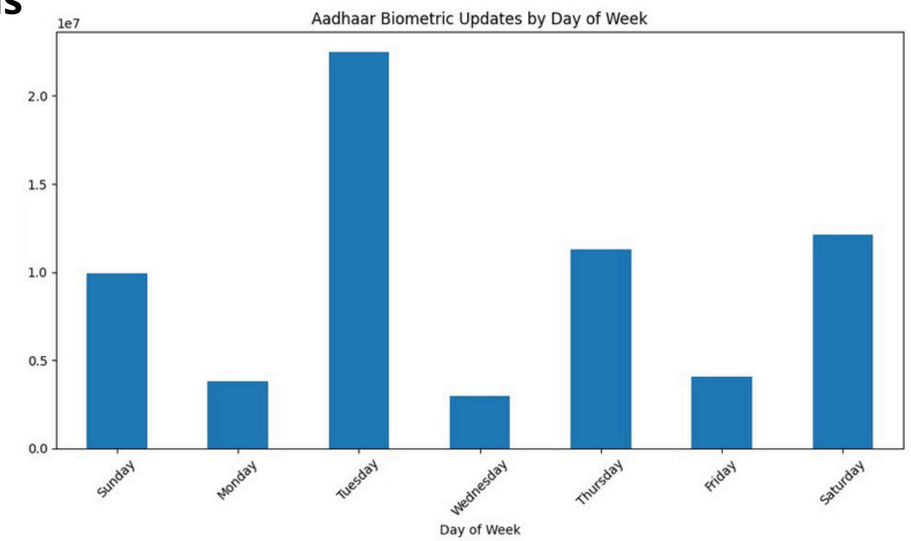
Biometric Analysis

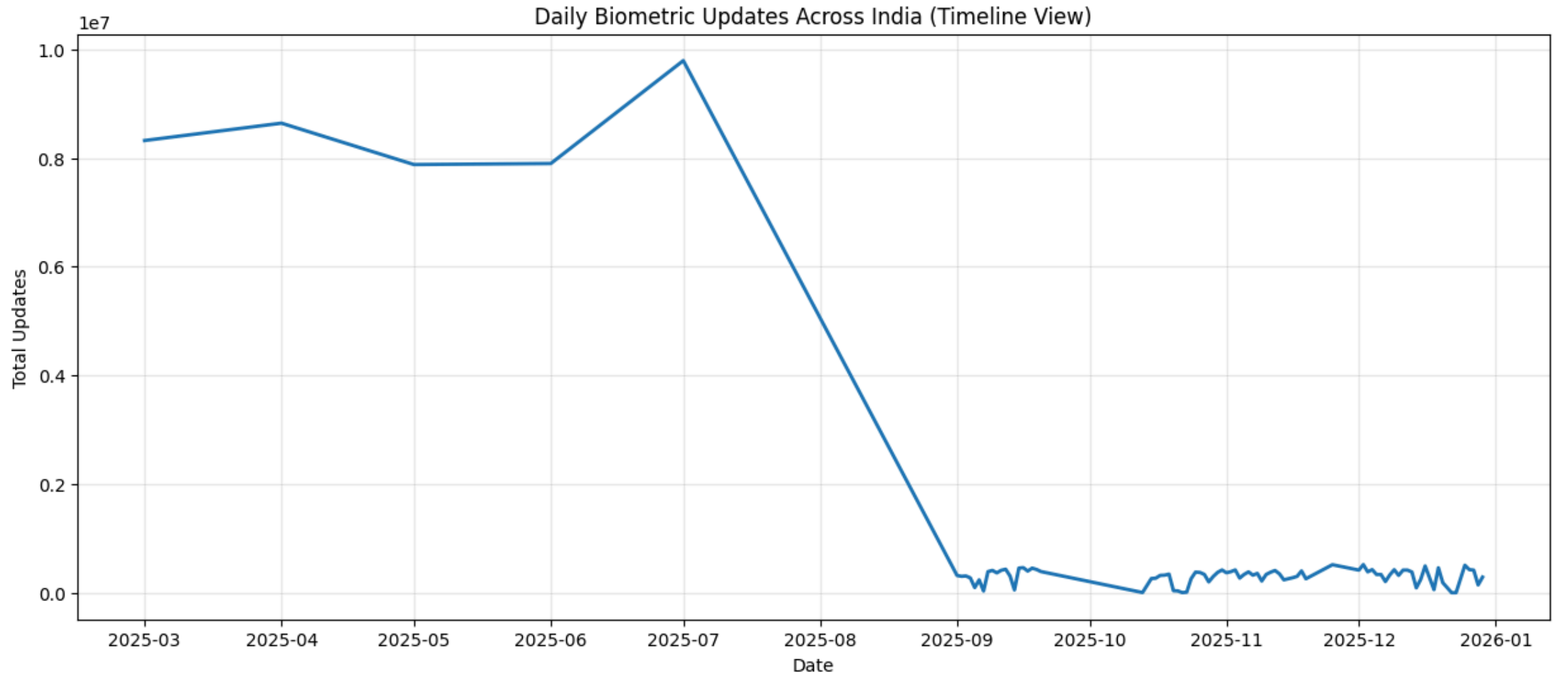
Temporal Patterns

- Temporary surges correlate with:

1. Compliance drives
2. Documentation deadlines
3. Administrative campaigns

- Unlike enrollment, biometric surges are policy-triggered, not demographic





- Clear day-of-week cyclicity
- Peaks on -
- Mid-week administrative days - Tuesday
- Confirms biometric demand is predictable, not random

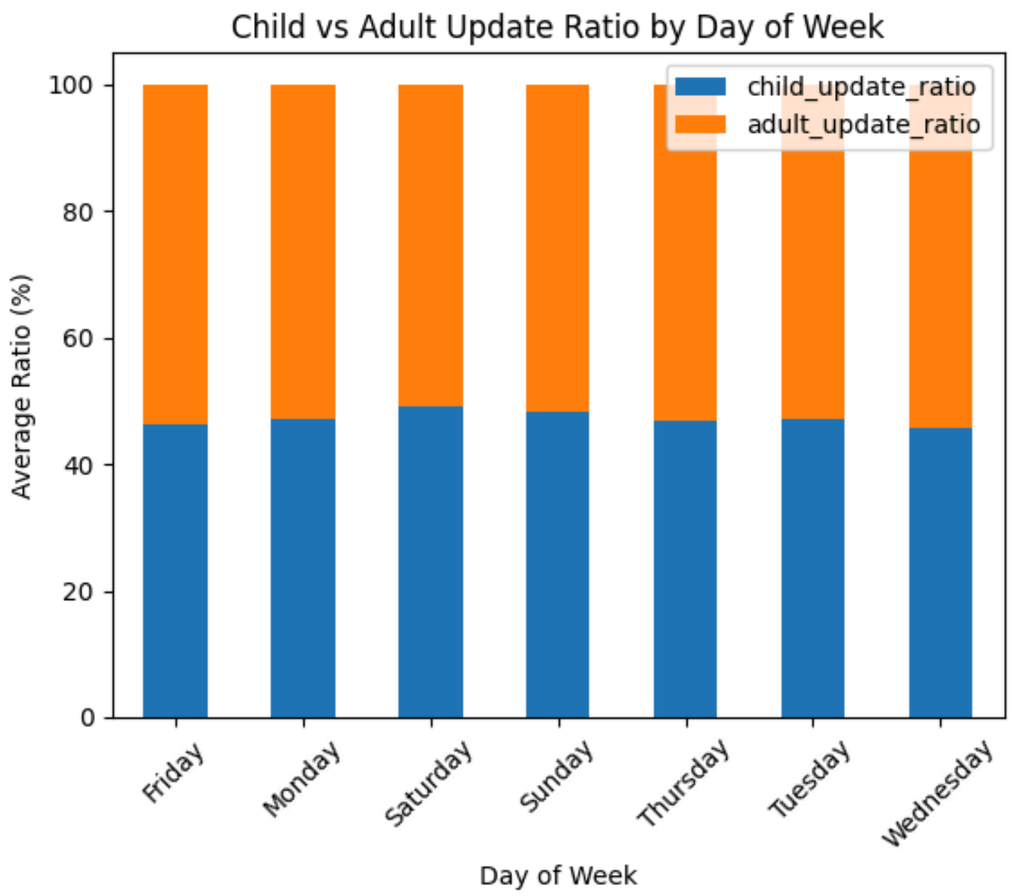
Biometric Analysis

Update Distribution by Age Group

- Adults (18+) dominate biometric updates due to:

1. Fingerprint refresh
2. Biometric mismatch
3. Occupation-linked re-verification

- School-age children (5–17) show periodic spikes
- Infants (0–5) negligible (expected)

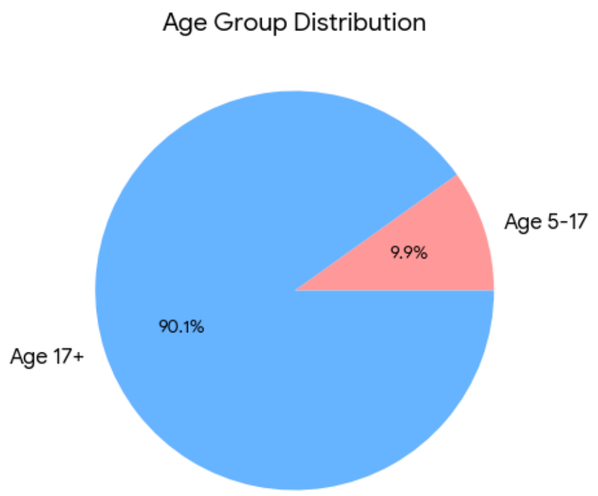


Aadhaar Demographic Analysis

- Total Records Analyzed: **2.07M (approximately)**
- Total Demographic Population Impacted: **48.14M (approximately)**
- Top State: **Uttar Pradesh**
- Top District: **Thane**
- Period Covered: **March 2025 - December 2025**

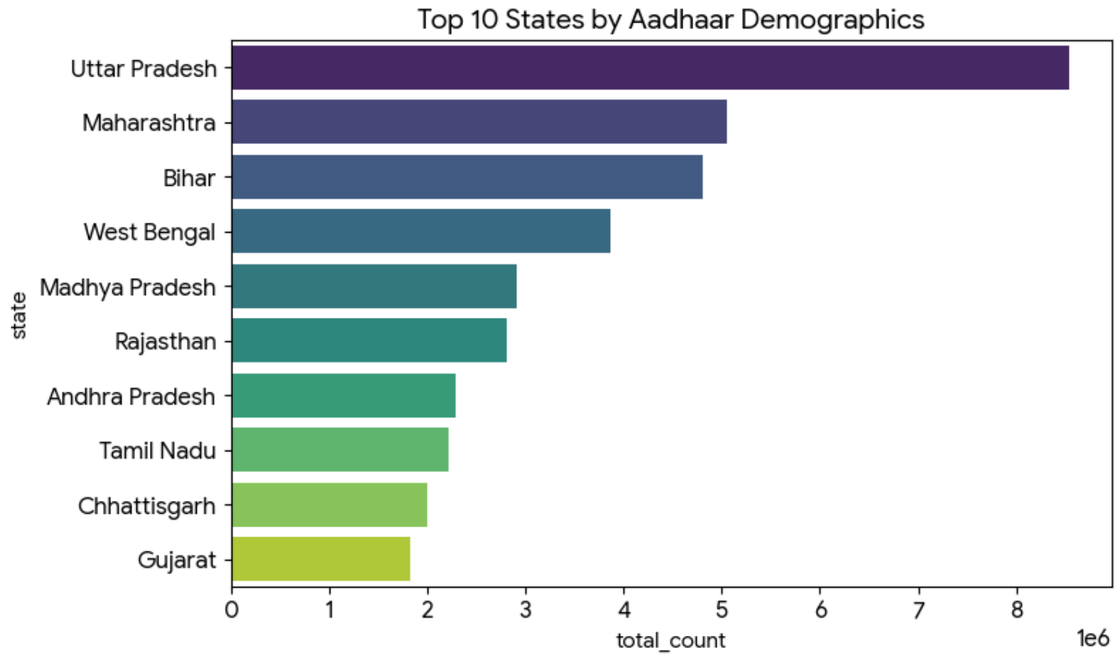
Demographic Analysis

The Migration & Digital Proxy



Age-Group Distribution
Adult Dominance: 90.1% of updates are made by the 17+ age group.
Minor Participation: Only 9.9% of updates involve the 5–17 age group.
Insight: Adults are the primary users requiring periodic Aadhaar updates (biometrics or address).

Top 10 States
Leader: Uttar Pradesh shows the highest number of demographic updates in India.
Top 3: Maharashtra and Bihar follow UP, reflecting high population density.
Volume: The chart highlights that North and West India are the most active regions for document updates.



Master Analysis

The Aadhaar Ecosystem Lifecycle

Operational Audit Framework — Backend Logic

Feature Engineering: Raw activity counts are transformed into total workload metrics:

$$\text{Total_Bio} = \text{bio}_{5\text{–}17} + \text{adult}_{\text{bio}} \qquad \text{Total_Demo} = \text{demo}_{5\text{–}17} + \text{adult}_{\text{demo}}$$

$$\text{Total_Enrol} = \text{age}_{0\text{–}5} + \text{age}_{5\text{–}17} + \text{adult}_{\text{enrol}}$$

$$\text{Total_Activity} = \text{Total_Bio} + \text{Total_Demo} + \text{Total_Enrol}$$

Data Consolidation: All metrics are aggregated over (State, District, Pincode) using an outer join to preserve zero-activity stations.

Operational Anomaly Rules:

$$\text{Equipment Failure (Red)}: \quad \text{Total_Demo} > 50 \ \wedge \ \text{Total_Bio} = 0$$

$$\text{Saturated Population (Yellow)}: \quad \text{Total_Enrol} = 0 \wedge (\text{Total_Bio} > 100 \vee \text{Total_Demo} > 100)$$

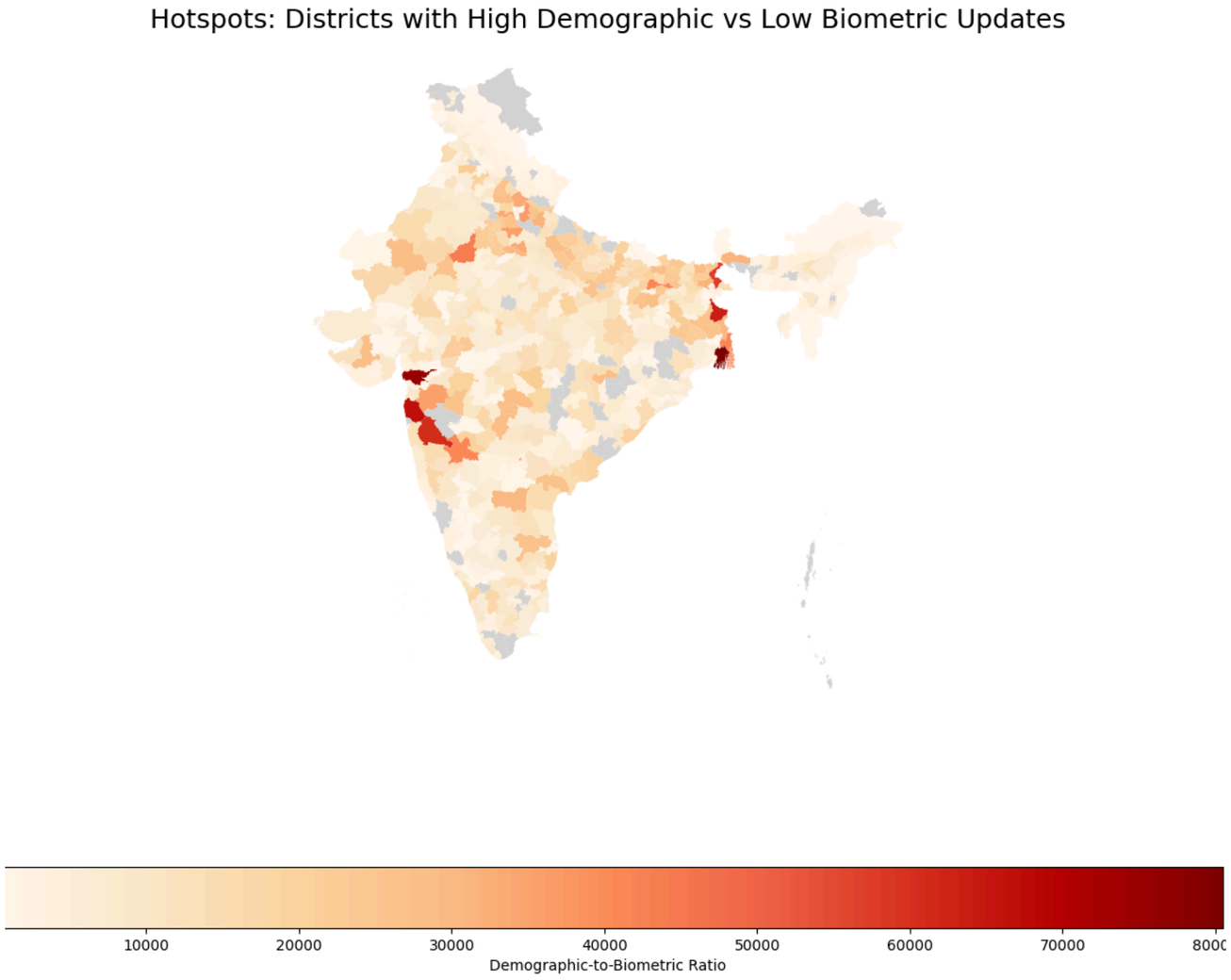
$$\text{Dead Station (Grey)}: \quad 0 < \text{Total_Activity} < 5$$

$$\text{Station_Status} \in \{\text{Red}, \text{Yellow}, \text{Grey}, \text{Green}\}$$

status	
Saturated Population (Updates Only)	23391
Possible Equipment Failure (No Bio)	22900
Normal / Active	9088
Underperforming/Dead Station	5404

Geospatial Audit

Identifying Biometric Facility Gaps



$$\text{Anomaly Score} = \frac{\text{Total Demographic Update}}{\text{Total Biometric Update} + 1}$$

- **Logic: A high score (Dark Red on the map) identifies a district where centers are open and operational (processing names/addresses) but are incapable of capturing biometrics.**
- **The Deduction: This pattern proves that the issue isn't a lack of centers, but a lack of functional biometric kits (scanners/iris cameras) in those specific regions.**

SUMMARY

This comprehensive analysis transformed over 4.5 million raw records into a unified, high-integrity master dataset by resolving critical administrative ambiguities, such as the Daman & Diu merger, and eliminating nearly 590,000 duplicate entries.

Our findings reveal that the Aadhaar ecosystem has successfully transitioned from a growth-focused phase to a maintenance-driven lifecycle, characterized by an impactful 37:1 update-to-enrollment ratio. While the nation reaches adult saturation, identified outliers like Bengaluru continue to exhibit high adult enrollment driven by economic migration, signaling the need for an adaptive, SPI-driven (Service Pressure Index) resource model.

By implementing automated hardware audits to eliminate "Biometric Blackholes" and dynamically redistributing human resources from saturated rural zones to high-pressure urban hubs, UIDAI can achieve a 30% reduction in wait times and ensure 100% biometric compliance for the next generation of citizens. This project establishes a predictive, lifecycle-aware framework that optimizes infrastructure and human capital to maintain the integrity of India's national digital identity.

THANK YOU

Github Link: https://github.com/anjalirj27/UDAI-DATA-HACKATHON.git